

E.G.S.PILLAY ENGINEERING COLLEGE (Autonomous)

Approved by AICTE, New Delhi| Affiliated to Anna University, Chennai Accredited by NAAC
with A++ Grade| Accredited by NBA T 1
NAGAPATTINAM–611002



B.E. Biomedical Engineering Final year - Seventh Semester

COURSE CODE	COURSENAME	CATEGORY	L	T	P	C	MAX.MARKS		
							CA	ES	TOTAL
THEORY COURSES									
2302BM701	AI/ML for biomedical engineers	PCC	3	0	0	3	40	60	100
2302BM702	Medical Ethics and standards	PCC	3	0	0	3	40	60	100
2302BM703	Hospital Automation	PCC	3	0	0	3	40	60	100
	Professional Elective V	PCC	3	0	0	3	40	60	100
	Professional Elective VI	PEC	3	0	0	3	40	60	100
	Open elective-III	OE	3	0	0	3	40	60	100
PRACTICAL COURSES									
2302BM751	Project work Phase I	EEC	0	0		4	40	60	100
2302BM752	Industrial Certification	EEC	0	0	0	1	40	60	100
TOTAL			18	0	0	23	420	480	900

L–Lecture| T–Tutorial |P–Practical| C–Credit| CA–Continuous Assessment| ES–End Semester

2302BM701	AI/ML FOR BIOMEDICAL ENGINEERS						L	T	P	C			
	(B.E-BME)						3	0	0	3			
PREREQUISITE:													
	Mathematical foundations, Programming skills, Biomedical domain knowledge, and Data-handling ability												
COURSE OBJECTIVES:													
	- To understand the problem-solving intelligent agents and searching techniques. - To Impart domain knowledge in different machine learning method. - To realize the different applications in AI												
COURSE OUTCOMES:													
On the successful completion of the course the students will be able to,													
CO1	Analyze the concepts of intelligent agents and its structure												
CO2	Apply appropriate search algorithms for solving given AI problems.												
CO3	Differentiate learning strategies, regression and classification in Artificial Intelligence Systems.												
CO4	Analyze the basic concepts of reinforcement learning and find solutions												
CO5	Apply the machine learning techniques in AI applications.												
COs Vs POs MAPPING:													
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
	CO1	2	1	2	-	2	2	-	-	-	-	-	1
	CO2	2	2	1	2	2	-	-	-	-	-	-	1
	CO3	2	2	2	3	2	-	-	-	-	-	-	1
	CO4	3	2	2	1	2	-	2	-	-	-	-	1
	CO5		2	2	-	2	-	-	-	-	-	-	2
COs Vs PSOs MAPPING													
	COs	PSO1	PSO2	PSO3									
	CO1	1	2	1									
	CO2	2	2	1									
	CO3	2	2	1									
	CO4	1	2	1									
	CO5	2	2	1									
COURSE CONTENTS:													
MODULE I	INTELLIGENT AGENTS										9 Hours		
Introduction to AI, Agents and Environments, Concept of rationality, Nature of environments, Structure of agents, problem solving agents, search algorithms, uninformed search strategies.													
MODULE II	PROBLEM SOLVING										9 Hours		
Heuristic search strategies, Heuristic functions, Local search and optimization problems, Local search in continuous space Online search agents and unknown environments, optimal Decisions in games, Constraint satisfaction problems (CSP).													
MODULE III	MACHINE LEARNING METHODS										9 Hours		
Forms of learning, Supervised learning, learning decision trees, Evaluation and choosing the best hypothesis, Theory of Learning, Regression and classification with linear models, Artificial Neural network, Non parametric model, Support vector machine, Ensemble learning.													
MODULE IV	REINFORCEMENT LEARNING										9 Hours		
Introduction to Reinforcement Learning, Active and Passive Reinforcement Learning, Generalization in reinforcement learning, Policy Search, Applications of Reinforcement Learning.													

MODULE V	AI APPLICATIONS	9 Hours
Natural Language Processing Language Models, Text Classification, Information Retrieval, Information Extraction, Machine Translation, Speech Recognition, Robotics, Robotic Hardware and Robotic perception.		
	TOTAL:	45 Hours
FURTHER READING: Broad Biomedical ML & Informatics		
Text books:		
1. Stuart Jonathan Russell, Peter Norvig, John Canny, Artificial Intelligence: A Modern Approach, Prentice Hall, Fourth edition, 2020		
2. Ameet V Joshi, Machine Learning and Artificial Intelligence, Springer Publications, 2020		
References:		
1. T.M. Mitchell, Machine Learning, McGraw-Hill 2017		
2. Ethem Alpaydin, Introduction to Machine Learning, MIT Press, Third edition 2014		
3. Stephen Marsland, Machine Learning An Algorithmic Perspective, Second Edition, Chapman and Hall/CRC Machine Learning and Pattern Recognition Series, 2014		

2302BM702		MEDICAL ETHICS AND STANDARDS								L	T	P	C	
		(B.E-BME)								3	0	0	3	
PREREQUISITE:														
		Human Anatomy & Physiology												
COURSE OBJECTIVES:														
		1.To introduce the legal and ethical principles in health care settings												
		2.To familiarize the professional ethics to be followed by Biomedical Engineers												
		3.To explore the patient safety and regulatory aspects followed in hospitals												
COURSE OUTCOMES:														
On the successful completion of the course the students will be able to,														
CO1		1.Implement legal and professional guidelines for the health professions												
CO2		2.Apply regulatory codes of ethics in healthcare systems												
CO3		3.Implement medical device safety aspects based on quality systems requirement												
CO4		4.Analyze success and failure aspects of bioethics												
CO5		5. Analyze the need of sustainable bioethics												
COs Vs POs MAPPING:														
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	
	CO1	2	1	2	-	2	2	-	-	-	-	-	1	
	CO2	2	2	1	2	2	-	-	-	-	-	-	1	
	CO3	2	2	2	3	2	-	-	-	-	-	-	1	
	CO4	3	2	2	1	2	-	2	-	-	-	-	1	
	CO5		2	2	-	2	-	-	-	-	-	-	2	
COs Vs PSOs MAPPING														
		COs	PSO1	PSO2	PSO3									
		CO1	1	2	1									
		CO2	2	2	1									
		CO3	2	2	1									
		CO4	1	2	1									
		CO5	2	2	1									
COURSE CONTENTS:														
MODULE I		INTRODUCTION TO MEDICAL ETHICS										9 Hours		
Definition of Medical ethics, Scope of ethics in medicine, International code of Ethics for occupational health professionals, Ethical Theories, Deontology & Utilitarianism, Casuisttheory, Virtuetheory, The Right Theory. Role of ethics in Healthcare workplace, Autonomy, Non Malfeasance, Beneficence, Veracity, Justice, OSHA, Decision Model for Healthcare Dilemmas, Applications of Plus decision making model.														
MODULE II		CODE OF ETHICS FOR BIOMEDICAL ENGINEERS										9 Hours		
Bioethics-The principle of Double effect, Code of Hammurabi, Engineering Competence, Ethical Issues in biomedical research Cloning and stem cell research, Neuro ethics, Organ Transplantation, Hypothetico deductive method, Research Conflict of Interest.														
MODULE III		MEDICAL DEVICE SAFETY										9 Hours		
Shared Responsibility for Medical device safety. WHO International Health Regulations (IHR),Stagof regulatory control of medical devices, Ethics committee, its members and functions, Global														

Harmonization Task Force (GHTF). Quality systems requirement, ISO, Voluntary and mandatory standards, Collateral Standards EMC radiation protection & programmable medical device system, Particular Standards-type of medical device.		
MODULE IV	BIO ETHICAL SUCCESS AND FAILURE	9 Hours
Measurements of success and Failure, Technological Success and failure, Risk as a bioethical concept, Safety, risk and reliability in design, Reliability An ethics metric, reducing risk, risk as an ethical concept, risk based ethics. Medical device failure, Five failure types, Bio-terrorism.		
Module V	SUSTAINABLE BIOETHICS	9 Hours
Introduction to Sustainable Bioethics Rational ethics, Life cycles and Concurrent Engineering, Bioethics of Combustion, Systematic Bioethics Seveso Plant disaster, Poverty and Pollution, Interdependence, Macro ethics and Micro ethics, The Humble Engineer		
TOTAL:		45 HOURS
FURTHER READING: Bio waste management		
Text books:		
1. William Charney, Hand book of Modern Hospital Safety, CRC Press, 2 nd Edition, 2009. 2. Almira Badnjevic, Mario Cifrek, Ratko Magjarevic, Zijad Dzemic , Inspection of Medical Devices: For Regulatory Purposes, Springer Nature, 2018.		
References:		
3. Daniela Vallero, Biomedical Ethics for Engineers, Elsevier Publications, 1 st Edition, 2007.		
4. Eileen E.Morrison, Ethics in Health Administration: A Practical Approach for Decision Makers, Jonnes and Bartletts Publication, 2nd Edition, 2011.		
5. Robert M Veatch, Basics of Bio Ethics, Prentice Hall,Inc.,2 nd Edition, 2003		

2303BM703		HOSPITAL AUTOMATION						L	T	P	C	
		(B.E-BME)						3	0	0	3	
PREREQUISITE:												
		Clinical engineering										
COURSE OBJECTIVES:												
		Introduce the concepts of hospital systems and need for central monitoring										
		Exemplify the power generation, utility and protection systems.										
		Apply the distributed and central monitoring functions in hospital environment										
COURSE OUTCOMES:												
On the successful completion of the course the students will be able to,												
CO1		Identify the factors in central power generating and monitoring systems										
CO2		Analyze the sensors and actuators for the automation systems										
CO3		Classify the equipment types and its applications.										
CO4		Apply software tools and digital computer for monitoring of parameters and medical data handling										
CO5		Design central monitoring station for hospitals for control and surveillance applications										
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2											
CO2		2										
CO3		2										
CO4		2										
CO5	3											
COs Vs PSOs MAPPING												
				COs	PSO1	PSO2	PSO3					
				CO1		2						
				CO2		2						
				CO3	3							
				CO4		2						
				CO5		2						
COURSE CONTENTS:												
MODULE I		AUTOMATION IN HEALTHCARE									9 Hours	
Introduction to automation Role of automation in healthcare Remote Patient Monitoring Maximizing resources on patient care Reducing variability, Automating clinician and patient interactions through products.												
MODULE II		POWER GENERATION AND MEDICAL GAS PRODUCTION									9 Hours	
Power generator, Battery : Maintenance and troubleshooting, energy conservation and monitoring system - Automation in dryer, compressor, air conditioning, lighting, heating systems.												
MODULE III		AUTOMATION IN PIPING									9 Hours	
Monitoring of flow and pressure of medical gas System components Vacuum control units Automatic changeover system - Types of Outlets - Leakage test- Prevention and safety automation.												
MODULE IV		INSTRUMENTATION SYSTEMS									9 Hours	
Optical sensors , Pressure Sensors - Ultrasonic Sensors - Tactile Sensors - Thermal sensors -Biosensor - Linear Actuators, Central monitoring station - Alarm system - Regulation and standards.												

MODULE V	APPLICATIONS	9 Hours
Business intelligence & executive dashboards - Radio-Frequency Identification (RFID)- based patient and asset tracking solutions - Tablet-based applications for bed side access to doctors/nurses - Healthcare CRM for patient relationship management - Patient kiosk, tele-health – HIS integration.		
TOTAL:		45 Hours
FURTHER READING: Safety Management		
References:		
1. Khandpur RS, Handbook of Biomedical Instrumentation, Prentice Hall of India, New Delhi, 3 rd edition, 2014.		
2. Joseph J. Carr and John M. Brown, Introduction to Biomedical Equipment Technology, Pearson Education India, Delhi, 4 th edition 2008		
3. Curtis Johnson D Process Control Instrumentation Technology, Prentice Hall of India, 8 th edition 2006		
4. John V. Grimaldi and Rollin H. Simonds., Safety Management, All India Travelers Book seller, New Delhi, 1989		

Professional Elective - V

SL. NO.	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1.	2303BM026	Principles of Tissue Engineering	PEC	3	0	0	3	3
2.	2303BM027	Medical Innovation and Entrepreneurship	PEC	3	0	0	3	3
3.	2303BM028	Bio statistics	PEC	3	0	0	3	3
4.	2303BM029	Ergonomics	PEC	3	0	0	3	3
5.	2303BM030	Medical Imaging Systems	PEC	3	0	0	3	3
6.	2303BM031	Tele health Technology	PEC	3	0	0	3	3

2303BM026		PRINCIPLES OF TISSUE ENGINEERING				L	T	P	C				
		(B.E-BME)				3	0	0	3				
PREREQUISITE:													
		Biomechanics											
COURSE OBJECTIVES:													
		1.To introduce the fundamentals and interdisciplinary aspects of tissue engineering.											
		2.To understand the role of cells, scaffolds, and signaling molecules in tissue regeneration.											
		3.To explore the design and application of biomaterials and bioreactors in tissue engineering.											
		4.To expose students to current clinical applications and ethical issues in regenerative medicine.											
COURSE OUTCOMES:													
On the successful completion of the course the students will be able to,													
CO1	Describe the components and basic principles of tissue engineering.												
CO2	Explain the types and functions of cells involved in tissue regeneration.												
CO3	Analyze the design and fabrication methods of scaffolds used in tissue engineering												
CO4	Evaluate tissue-specific engineering approaches and bioreactor usage.												
CO5	Discuss the clinical applications, ethical, and regulatory issues in tissue engineering.												
COs Vs POs MAPPING:													
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
	CO1	2	2	1				1					
	CO2	2	2					1	1				
	CO3	2	2	3	2	2				1			
	CO4	2	2	2	3	2		1					
	CO5	2	2				2	3	3				1
COs Vs PSOs MAPPING													
		COs	PSO1	PSO2	PSO3								
		CO1	2	1	2								
		CO2	2	1	2								
		CO3	2	1	2								
		CO4	2	2	2								
		CO5	2	2	2								
COURSE CONTENTS:													
MODULE I		INTRODUCTION TO TISSUE ENGINEERING								9 Hours			
Definition, scope, and interdisciplinary nature of tissue engineering. Basic components – Cells, scaffolds, and signaling molecules. History and evolution of tissue engineering. Overview of regenerative medicine and clinical need. Challenges in tissue regeneration.													
MODULE II		CELLS IN TISSUE ENGINEERING								9 Hours			
Cell types: Stem cells (embryonic, adult, iPSCs), somatic cells. Cell sources, isolation, culture, and expansion techniques. Cellular microenvironment and extracellular matrix (ECM).Cell signaling and													

differentiation. Cell–scaffold interactions and tracking.		
MODULE III	BIOMATERIALS AND SCAFFOLD DESIGN	9 Hours
Introduction to biomaterials – natural and synthetic. Scaffold properties: Biocompatibility, porosity, mechanical strength. Scaffold fabrication methods: Electro spinning, solvent casting, 3D printing. Surface modification techniques. Degradation and integration with host tissue.		
MODULE IV	TISSUE ENGINEERING STRATEGIES	9 Hours
Tissue-specific strategies: Bone, cartilage, skin, blood vessels, nerve tissue. Role of bioreactors – types and applications. Growth factors and controlled release systems. Vascularization and immune responses. In vitro and in vivo models for tissue testing.		
MODULE V	CLINICAL APPLICATIONS AND ETHICAL ISSUES	9 Hours
Case studies: Engineered skin, cornea, trachea, urinary bladder. Regulatory guidelines and translational challenges (FDA, CDSCO). Preclinical and clinical trial considerations. Commercialization aspects. Ethical and societal issues in regenerative therapies		
		TOTAL: 45 Hours
FURTHER READING: Hospital training		
Text books:		
1.Lanza, R., Langer, R., & Vacanti, J. P. <i>Principles of Tissue Engineering</i> , 4th Ed., Academic Press, 2013.		
2.Palsson, B., & Bhatia, S. <i>Tissue Engineering</i> , Pearson Education, 2004.		
References:		
1.Bernhard O. Palsson, <i>Introduction to Tissue Engineering</i> , Cambridge University Press, 2012.		
2.Clemens Van Blitterswijk, <i>Tissue Engineering</i> , Academic Press, 2008		

2303BM027	MEDICAL INNOVATION AND ENTREPRENEURSHIP					L	T	P	C			
	(B.E-BME)					3	0	0	3			
PREREQUISITE:												
	Clinical engineering											
COURSE OBJECTIVES:												
	1.To introduce the fundamentals of innovation and entrepreneurship in the biomedical field.											
	2.To enable students to identify unmet medical needs and propose innovative solutions.											
	3.To equip students with knowledge of product development, intellectual property, and regulatory affairs.											
	4.To provide an understanding of startup creation, funding, and business models in healthcare.											
COURSE OUTCOMES:												
On the successful completion of the course the students will be able to,												
CO1	Understand the fundamentals of innovation and entrepreneurship in biomedical engineering.											
CO2	Identify and assess unmet clinical needs and convert them into viable product ideas.											
CO3	Apply knowledge of medical device design, prototyping, and validation.											
CO4	Analyze intellectual property rights, regulatory pathways, and ethical considerations.											
CO5	Develop a business model and plan for a healthcare-based startup											
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1								1		
CO2	2	2	2	2			1	1	1			
CO3	2	2	3	2	2							
CO4	1	2	2			2	3	3		1		1
CO5	2	2	2		1	2			3	3		2
COs Vs PSOs MAPPING												
				COs	PSO1	PSO2	PSO3					
				CO1	1	2	2					
				CO2	2	2	2					
				CO3	2	2	2					
				CO4	2	2	2					
				CO5	2	3	3					
COURSE CONTENTS:												
MODULE I		INTRODUCTION TO MEDICAL INNOVATION							9 Hours			
Innovation in healthcare – historical perspective and current landscape. Need identification in clinical settings. Design thinking process in medical innovation. Overview of the medical product development lifecycle. Success and failure case studies in medical innovation.												
MODULE II		MEDICAL DEVICE DESIGN AND PROTOTYPING							9 Hours			

Problem selection and user-centric design Concept development, prototyping techniques (low- and high-fidelity). Usability and safety considerations in design. Introduction to rapid prototyping and 3D printing. Preclinical testing and validation of medical devices.		
MODULE III	INTELLECTUAL PROPERTY AND REGULATIONS	9 Hours
Basics of intellectual property: Patents, trademarks, copyrights. Patent filing process – national and international. Medical device classification and regulatory bodies (FDA, CDSCO, CE). Standards and certifications (ISO 13485, IEC 60601). Ethical and legal aspects in innovation.		
MODULE IV	ENTREPRENEURSHIP AND BUSINESS MODELS	9 Hours
Entrepreneurial mindset and qualities. Business model canvas and value proposition design. Startup lifecycle: Ideation to commercialization. Types of business entities and legal considerations. Case studies of Indian and global Med Tech startups.		
MODULE V	FUNDING AND GO-TO-MARKET STRATEGY	9 Hours
Sources of funding: Grants, angel investors, venture capital, crowd funding. Pitching techniques and investor presentations. Product-market fit and market analysis. Sales, marketing, and distribution strategies for healthcare. Incubation, acceleration, and government initiatives (e.g., BIRAC, Atal Incubators).		
		TOTAL: 45 Hours
FURTHER READING: Hospital automation		
Text books:		
1. Paul Yock, Stefanos Zenios et al., <i>Bio design: The Process of Innovating Medical Technologies</i> , Cambridge University Press, 2015.		
2. Krishnamurthy, R., <i>Introduction to Medical Entrepreneurship</i> , Med Tech Publishers, 2020.		
References:		
1. Marc S. Levine, <i>The Lean Entrepreneur: Biomedical Edition</i> , Wiley, 2021.		
2. John Nesheim, <i>High Tech Startup: The Complete Handbook for Creating Successful New High Tech Companies</i> , Free Press, 2000.		
3. Government of India – <i>Startup India & BIRAC Guidelines</i> , official portals		

2303BM028	BIO STATISTICS								L	T	P	C
	(B.E-BME)								3	0	0	3
PREREQUISITE:												
	Hospital training											
COURSE OBJECTIVES:												
	1.To introduce statistical concepts relevant to biology and medicine.											
	2.To enable data interpretation using descriptive and inferential statistics.											
	3.To apply statistical tools for clinical and experimental biomedical data.											
	4.To develop skills in hypothesis testing, correlation, regression, and probability.											
COURSE OUTCOMES:												
On the successful completion of the course the students will be able to,												
CO1	Describe basic statistical terms and summarize biomedical data using descriptive statistics.											
CO2	Apply probability concepts and distributions to interpret biological data.											
CO3	Perform hypothesis testing using parametric and non-parametric tests.											
CO4	Use correlation and regression methods to analyze experimental data											
CO5	Interpret biomedical research data using statistical software tools											
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2								1		
CO2	3	2	2									
CO3	3	3	2	1								
CO4	3	3	3	2								
CO5	2	3	2	2	2							1
COs Vs PSOs MAPPING												
				COs	PSO1	PSO2	PSO3					
				CO1	1	1	2					
				CO2	1	2	2					
				CO3	2	2	2					
				CO4	2	2	2					
				CO5	2	2	3					
COURSE CONTENTS:												
MODULE I		INTRODUCTION TO BIOSTATISTICS									9 Hours	
Definition and scope of biostatistics. Types of data: Qualitative and quantitative. Measurement scales: Nominal, ordinal, interval, and ratio. Collection, classification, and tabulation of data. Diagrammatic and graphical representation: Bar diagram, histogram, pie chart.												
MODULE II		MEASURES OF CENTRAL TENDENCY AND DISPERSION									9 Hours	
Mean, median, mode – definitions and calculation. Measures of dispersion: Range, mean deviation, variance, standard deviation, coefficient of variation. Skewness and kurtosis – interpretation and application in biomedical data. Applications in epidemiology and hospital data.												
MODULE III		PROBABILITY AND DISTRIBUTIONS									9 Hours	

Basic probability rules and concepts. Bayes' theorem and its biomedical relevance. Probability distributions: Binomial, Poisson, and Normal distribution. Standard scores (Z-scores) and applications. Central Limit Theorem.		
MODULE IV	HYPOTHESIS TESTING	9 Hours
Concept of hypothesis – null and alternative. Types of errors, level of significance, and power of the test. Parametric tests: t-test (one sample, two sample), z-test, ANOVA. Non-parametric tests: Chi-square test, Mann–Whitney U test. Application in clinical trial data analysis.		
MODULE V	CORRELATION, REGRESSION AND SOFTWARE TOOLS	9 Hours
Correlation: Pearson and Spearman methods. Regression analysis: Simple linear and multiple regression. Introduction to statistical software: SPSS, R, Excel. Design of experiments and data interpretation. Real-life case studies using biomedical datasets.		
		TOTAL: 45 HOURS
FURTHER READING: Medical coding		
Text books:		
1. Wayne W. Daniel, <i>Biostatistics: A Foundation for Analysis in the Health Sciences</i> , Wiley, 10th Ed., 2013.		
2. Khan, I. A., &Khanum, A., <i>Fundamentals of Biostatistics</i> , Ukaaz Publications, 2012.		
References:		
1.Sundar Rao, <i>Introduction to Biostatistics and Research Methods</i> , PHI Learning, 2010.		
2.Arora, P. N., &Malhan, P. K., <i>Biostatistics</i> , Himalaya Publishing House, 2013.		
3.Rosner, B., <i>Fundamentals of Biostatistics</i> , Cengage Learning, 2015.		

2303BM029	ERGONOMICS		L	T	P	C						
	(B.E-BME)		3	0	0	3						
PREREQUISITE:												
	Human anatomy,Biomechanics											
COURSE OBJECTIVES:												
	1.To introduce the principles and significance of ergonomics in healthcare and biomedical design.											
	2.To understand human anatomical, physiological, and cognitive characteristics relevant to design											
	3.To apply ergonomic principles to the design of medical devices, workplaces, and rehabilitation tools.											
	4.To evaluate ergonomic risks and suggest improvements for safety and efficiency											
COURSE OUTCOMES:												
On the successful completion of the course the students will be able to,												
CO1	Explain the fundamentals of ergonomics and its role in biomedical engineering.											
CO2	Analyze human anatomical and physiological factors affecting workplace and equipment design.											
CO3	Apply ergonomic principles in the design and evaluation of medical and assistive devices.											
CO4	Assess and manage ergonomic risks in clinical and industrial environments.											
CO5	Integrate ergonomics with product development, rehabilitation, and patient care solutions.											
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2					1			1	1	1
CO2	2	2	2								1	1
CO3	2	3	3	2	2		1				1	1
CO4	2	3	2	2			3	2		1	1	1
CO5	2	2	2	2	2	1			1	1	1	1
COs Vs PSOs MAPPING												
COURSE CONTENTS:												
MODULE I		INTRODUCTION TO ERGONOMICS									9 Hours	
Definition and scope of ergonomics. Importance in biomedical and clinical environments. Classification: Physical, cognitive, and organizational ergonomics. Human-machine interaction and systems approach. Historical development and applications in medical settings.												

MODULE II	HUMAN ANATOMY AND PHYSIOLOGY IN ERGONOMICS	9 Hours
Musculoskeletal system and anthropometry. Static and dynamic posture analysis. Work physiology and metabolic rate. Biomechanics and manual material handling. Physiological factors in rehabilitation device design.		
MODULE III	ERGONOMIC DESIGN PRINCIPLES	9 Hours
Anthropometric data and workspace design. Ergonomic principles in tool and device design. Display and control design – visual, auditory, tactile. Environmental factors – lighting, noise, vibration, and thermal comfort. Design of hospital furniture and patient support systems.		
MODULE IV	COGNITIVE AND ORGANIZATIONAL ERGONOMICS	9 Hours
Cognitive load and human error. Decision-making, perception, memory, and stress. Human–computer interaction and usability. Teamwork, work-rest cycles, and shift scheduling. Human reliability and mental workload assessment.		
MODULE V	ERGONOMIC RISK ASSESSMENT AND APPLICATIONS	9 Hours
Ergonomic assessment methods: RULA, REBA, NIOSH, OWAS. Workplace safety analysis and accident prevention. Assistive device ergonomics – wheelchairs, orthotics, prosthetics. Ergonomics in telemedicine and wearable technology. Case studies: Surgical ergonomics, ICU design, rehabilitation settings		
		TOTAL: 45 Hours
FURTHER READING: Occupational Ergonomics		
Text books:		
1.Bridger, R. S., <i>Introduction to Ergonomics</i> , 3rd Ed., CRC Press, 2009.		
2.Kroemer, K. H. E., &Grandjean, E., <i>Fitting the Task to the Human</i> , Taylor & Francis, 2001.		
References:		
1.Dul, J., &Weerdmeester, B., <i>Ergonomics for Beginners: A Quick Reference Guide</i> , CRC Press, 2008		
2.Konz, S. A., & Johnson, S., <i>Work Design: Occupational Ergonomics</i> , Holcomb Hathaway, 2007.		
3.Sanders, M. S., & McCormick, E. J., <i>Human Factors in Engineering and Design</i> , McGraw-Hill2013		

2303BM030		MEDICAL IMAGING SYSTEMS				L	T	P	C					
		(B.E-BME)				3	0	0	3					
PREREQUISITE:														
		Image processing												
COURSE OBJECTIVES:														
		- To understand the principles of various medical imaging modalities. - To explore image formation, instrumentation, and data acquisition techniques. - To analyze the advantages and limitations of different imaging systems. - To familiarize with clinical applications of imaging technologies.												
COURSE OUTCOMES:														
On the successful completion of the course the students will be able to,														
CO1		Explain the physics and working principles behind various medical imaging modalities.												
CO2		Analyze the components and functioning of imaging instruments and systems.												
CO3		Apply signal and image processing techniques in medical image acquisition and enhancement.												
CO4		Evaluate the clinical relevance and limitations of imaging technologies.												
CO5		Compare various imaging systems based on resolution, safety, and application.												
COs Vs POs MAPPING:														
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	
	CO1	3	2									1		
	CO2	3	2	2										
	CO3	2	3	3	2	2								
	CO4	2	3	2	2		1	2				1	1	
	CO5	2	2	2						1		2	2	
COs Vs PSOs MAPPING														
COURSE CONTENTS:														
MODULE I		INTRODUCTION TO MEDICAL IMAGING										9 Hours		
Overview of medical imaging systems. Interaction of radiation with matter.														
Basic concepts of imaging physics – image resolution, contrast, signal-to-noise ratio. Classification of imaging modalities – ionizing vs non-ionizing. Patient safety, radiation dose, and biological effects.														
MODULE II		X-RAY AND COMPUTED TOMOGRAPHY (CT)										9 Hours		
X-ray production, attenuation, and image formation. X-ray machine components: Tube, collimators, detectors.Digital radiography and fluoroscopy. CT principles, generation, image reconstruction														

algorithms.CT detectors and helical CT. Applications and safety measures.		
MODULE III	ULTRASOUND IMAGING	9 Hours
Ultrasound physics: Acoustic impedance, reflection, attenuation. Transducers – types and working. A-mode, B-mode, M-mode, Doppler imaging.Real-time imaging and 3D/4D ultrasound. Clinical applications: Obstetrics, cardiology, musculoskeletal imaging.		
MODULE IV	MRI AND NUCLEAR IMAGING	9 Hours
MRI principles: Nuclear magnetic resonance, relaxation times (T1, T2).Magnet system, gradient coils, RF coils.Pulse sequences and MR image formation. Nuclear imaging: SPECT, PET – tracer principles, detectors. Clinical applications and safety aspects		
MODULE V	IMAGE PROCESSING AND FUTURE TRENDS	9 Hours
Basics of medical image processing: Enhancement, segmentation, filtering. DICOM standards and PACS.Machine learning and AI in medical imaging. Fusion imaging (PET-CT, PET-MRI).Future directions: Molecular imaging, optical imaging.		
TOTAL:		45 Hours
FURTHER READING: Hospital visit for MRI SCAN		
Text books:		
1. Bushberg, J. T., Seibert, J. A., Leidholdt, E. M., & Boone, J. M., <i>The Essential Physics of Medical Imaging</i> , Lippincott Williams & Wilkins, 2012.		
2. Suetens, P., <i>Fundamentals of Medical Imaging</i> , Cambridge University Press, 2017.		
References:		
1..Rangaraj M. Rangayyan, <i>Biomedical Image Analysis</i> , CRC Press, 2005.		
2.Hendee, W. R., &Ritenour, E. R., <i>Medical Imaging Physics</i> , Wiley, 2002.		
3.Khandpur, R. S., <i>Handbook of Biomedical Instrumentation</i> , Tata McGraw-Hill, 2014		

2303BM031	TELEHEALTH TECHNOLOGY	L	T	P	C
	(B.E-BME)	3	0	0	3

PREREQUISITE:

COURSE OBJECTIVES:

	1.To introduce the concepts and scope of telehealth and telemedicine systems.
	2.To understand the technologies and communication protocols used in telehealth delivery
	3.To explore clinical applications and integration of biomedical devices in telehealth platforms
	4.To understand legal, ethical, and regulatory issues involved in telemedicine

COURSE OUTCOMES:

On the successful completion of the course the students will be able to,

CO1	Explain the concepts and components of telehealth and telemedicine systems.
CO2	Identify appropriate communication technologies and standards used in telemedicine.
CO3	Integrate biomedical sensors and devices into telehealth platforms.
CO4	Evaluate data security, legal, and regulatory issues in telehealth systems.
CO5	Analyze applications of telehealth in clinical and rural healthcare delivery.

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2									1	1
CO2	2	3	2	2	2						1	1
CO3	2	3	3	2	2		1				1	1
CO4	2	2	2		2	3	2	2			1	1
CO5	2	2	2			2	3	3	1		1	1

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO3
CO1	2	2	2
CO2	2	2	2
CO3	2	2	2
CO4	2	2	3
CO5	2	3	3

COURSE CONTENTS:

MODULE I	INTRODUCTION TO TELEHEALTH AND TELEMEDICINE	9 Hours
Definition, evolution, and scope of telehealth. Difference between telehealth, telemedicine, Health, and eHealth. Telemedicine system architecture – types, modes, and applications. Benefits and limitations of telemedicine. Key stakeholders: patients, providers, technology partners.		
MODULE II	TELECOMMUNICATION INFRASTRUCTURE	9 Hours
Communication systems used in telemedicine – satellite, internet, wireless. Protocols and standards: HL7, DICOM, TCP/IP. Network types: LAN, WAN, VPN in tele health. Bandwidth, latency, and QoS issues in remote monitoring. Video conferencing and tele consultation tools.		
MODULE III	BIOMEDICAL DEVICES AND TELEMONTORING	9 Hours

Remote patient monitoring systems. Integration of biomedical sensors (ECG, BP, SpO ₂ , glucose, etc.). Wearables and IoT in tele health. Real-time vs store-and-forward methods. Data acquisition, transmission, and analysis.		
MODULE IV	DATA SECURITY AND REGULATORY COMPLIANCE	9 Hours
Data privacy and security in telemedicine – encryption, authentication. HIPAA, GDPR, and Indian IT Act. Medical data standards and inter operability. Risk management and cyber security in telehealth. Ethical and legal aspects of remote care delivery.		
MODULE V	CLINICAL APPLICATIONS AND INNOVATIONS	9 Hours
Tele consultation, tele radiology, tele cardiology, tele psychiatry. Mobile health apps and smart health care. Tele-ICU and rural health access Case studies of successful telehealth implementations (India & Global). Future trends: AI, robotics, block chain in telehealth.		
TOTAL:		45 Hours
FURTHER READING: Computer networks		
Text books:		
1. Norris, A. C., <i>Essentials of Telemedicine and Telecare</i> , Wiley, 2002.		
2. Bashshur, R., Shannon, G., & Krupinski, E., <i>Telemedicine: Theory and Practice</i> , Springer, 2015.		
References:		
3. Olga Ferrer-Roca and M. Sosa-Iudicissa, <i>Handbook of Telemedicine</i> , IOS Press, 2002.		
4. Khandpur, R. S., <i>Telemedicine Technology and Applications</i> , BS Publications, 2017.		
5. Mohan Bansal, <i>Introduction to Telemedicine</i> , Jaypee Brothers, 2021.		

Professional Elective - VI

SL. NO.	COURSE CODE	COURSE TITLE	CATEGORY	PERIODS PER WEEK			TOTAL CONTACT PERIODS	CREDITS
				L	T	P		
1.	2303BM032	Genetic Engineering	PEC	3	0	0	3	3
2.	2303BM033	Rapid Prototyping	PEC	3	0	0	3	3
3.	2303BM034	Forensic Science in Healthcare	PEC	3	0	0	3	3
4.	2303BM035	Haptics	PEC	3	0	0	3	3
5.	2303BM036	Computer Vision	PEC	3	0	0	3	3
6.	2303BM037	Medical Informatics	PEC	3	0	0	3	3

2303BM032	GENETIC ENGINEERING					L	T	P	C			
	(B.E-BME)					3	0	0	3			
PREREQUISITE:												
	Bio chemistry											
COURSE OBJECTIVES:												
	1.To understand the concepts of Genetics 2.To introduce the practice of recombinant DNA technologies 3.To solve genetic engineering problems and design target gene expression with advanced genetic engineering techniques. 4.To explore with genetic engineering techniques for cloning target gene or protein expression.											
COURSE OUTCOMES:												
On the successful completion of the course the students will be able to,												
CO1	Understand the concepts of Genetics											
CO2	Apply Recombinant DNA technology for gene cloning methods for construction of Gene Libraries											
CO3	Apply the polymer chain reaction in disease diagnosis, forensic science and genetic Engineering											
CO4	Analyse the advancements in genetic Engineering											
CO5	Analyse genetic engineering concepts in drugs and vaccine preparation and its ethical issues.											
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										
CO2	1	2	3			1		1				
CO3	1	2	3			1		1				
CO4	1	3	2			1		1				
CO5	1	3	2			1		3				
COs Vs PSOs MAPPING												
COs	PSO1	PSO2	PSO3									
CO1	3											
CO2	2	2										
CO3	2	2										
CO4	2	3										
CO5	2	3										
COURSE CONTENTS:												
MODULE I	BASICS OF GENETICS										9 Hours	
Biomolecules: Carbohydrates, Proteins, Lipid, Amino acid and Nucleic acids. Nucleic acids: Introduction, History, DNA and RNA- genetic material, types, mutation. Chromosome, Gene.												

Expression of genetic information, Regulation of mRNA stability.		
MODULE II	RECOMBINANT DNA TECHNOLOGY	9 Hours
Gene cloning - concept and basic steps; Restriction modification enzymes used in recombinant DNA technology, endonucleases, ligases and other enzymes useful in gene cloning; Vectors: plasmid, bacteriophage and other viral vectors, cosmids, artificial chromosomes, Ti plasmid, shuttle vectors, expression vectors; DNA delivery methods; Construction of genomic and cDNA libraries; Techniques for selection, screening and characterization of transformants.		
MODULE III	ROLE OF POLYMERASE CHAIN REACTION	9 Hours
Concept of PCR; DNA polymerases; primer designing, linkers, adapters, setting up PCR reactions; Various types of PCR; Applications of PCR in disease diagnostics, forensic sciences and genetic engineering.		
MODULE IV	ADVANCED APPROACHES IN GENETIC ENGINEERING	9 Hours
Gene expression in prokaryotes & eukaryotes, Tissue specific promoter, wound inducible promoters, Strong and regulatable promoters, promoter analysis (EMSA and DNA foot printing), gene expression profiling (real time PCR, SAGE, differential display, Microarray); DNA sequencing methods; Molecular markers: RAPD, RFLP, AFLP, SNP; Site directed mutagenesis, gene silencing techniques.		
MODULE V	APPLICATIONS OF GENETIC ENGINEERING	9 Hours
Genetic engineering and Biotechnology; Creation of recombinant microorganisms, transgenic plants and animals; cloning of sheep (Dolly) & other mammals; applications in conservation; therapeutic vs reproductive cloning; ethical issues and the prospects for human cloning; Gene therapy; DNA drugs and vaccines.		
TOTAL:		45 Hours
FURTHER READING: Visit Biopharm Industry		
Text books:		
1. Patrick Faraday, "Genetic Engineering: Emerging concepts and Technologies", Syrawood Publishers, 2018		
2. "The Biotech Primer: An Insider's Guide to the Science Driving the Biopharm Industry", The Biotech Primer For Non-Scientists Series, November 15, 2019		
References:		
1. Sandhya Mitra, "Genetic Engineering", McGraw Hill, 2nd edition, 2017.		
2. R. Wold, "Principles of gene manipulation - An introduction to genetic engineering", Distributors, USA, Publishers Business Services, 1989.		
3. Desmond S.T. Nicholl, "An Introduction to Genetic Engineering", Cambridge University Press, 2023.		

2303BM033		RAPID PROTOTYPING						L	T	P	C	
		(B.E-BME)						3	0	0	3	
PREREQUISITE:												
	Data structure and oops,Python											
COURSE OBJECTIVES:												
	1.To introduce the fundamentals and need for rapid prototyping in biomedical design.											
	2.To provide an understanding of various additive manufacturing techniques.											
	3.To enable students to convert design concepts into working biomedical prototypes.											
	4.To explore applications of rapid prototyping in personalized healthcare and implants.											
COURSE OUTCOMES:												
On the successful completion of the course the students will be able to,												
CO1	Understand the principles and process flow of rapid prototyping systems.											
CO2	Compare different additive manufacturing techniques and materials.											
CO3	Apply CAD modeling and slicing techniques for prototype generation.											
CO4	Evaluate post-processing methods and prototype validation strategies.											
CO5	Analyze biomedical applications of rapid prototyping and 3D printing.											
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2									1	1
CO2	3	2	2								1	1
CO3	2	3	3	2	3						1	1
CO4	2	2	2	2							1	1
CO5	2	2	2			2	2	2	1		1	1
COs Vs PSOs MAPPING												
				COs	PSO1	PSO2	PSO3					
				CO1	2	1	2					
				CO2	2	2	2					
				CO3	2	2	2					
				CO4	2	2	2					
				CO5	2	3	3					
COURSE CONTENTS:												
MODULE I		INTRODUCTION TO RAPID PROTOTYPING									9 Hours	
Definition, need and benefits of rapid prototyping in biomedical engineering. Comparison with traditional manufacturing. Process chain: Design – STL conversion – Slicing – Building – Post-processing.												
Applications in pre-surgical planning, prosthetics, dental models.												
MODULE II		ADDITIVE MANUFACTURING TECHNIQUES									9 Hours	
Overview of additive manufacturing methods: Stereo lithography (SLA) Fused Deposition Modeling (FDM)Selective Laser Sintering (SLS) Poly Jet, Inkjet printing Material extrusion and binder jetting. Advantages and limitations of each method												
MODULE III		CAD MODELING AND PRE-PROCESSING									9 Hours	

Introduction to 3D CAD tools: Solid Works, AutoCAD, Fusion 360.STL file generation and error correction. Slicing software – layer generation, orientation, support structures. G-code generation and printing parameters. Hands-on workflow (demo/project-based).		
MODULE IV	POST-PROCESSING AND VALIDATION	9 Hours
Support removal, surface finishing, curing and annealing techniques. Testing dimensional accuracy, surface roughness and structural properties. Mechanical and biological compatibility testing. Sterilization techniques for biomedical use. Documentation and quality assurance in prototyping labs.		
MODULE V	BIOMEDICAL APPLICATION	9 Hours
Tissue scaffolds and organ models using 3D printing. Custom implants, prosthetics, orthotic devices. Bio-printing basics – bio-inks, cell viability, layering strategies. Surgical simulation and training models. Recent trends: 4D printing, patient-specific devices.		
TOTAL:		45 Hours
FURTHER READING: Additive Manufacturing		
Text books:		
1. Chua, C. K., Leong, K. F., & Lim, C. S., <i>Rapid Prototyping: Principles and Applications</i> , World Scientific, 2010.		
2. Gibson, I., Rosen, D., & Stucker, B., <i>Additive Manufacturing Technologies</i> , Springer, 2015.		
References:		
3. Rafiq Noorani, <i>Rapid Prototyping: Principles and Applications in Manufacturing</i> , Wiley, 2005.		
4. Andreas Gebhardt, <i>Understanding Additive Manufacturing</i> , Hanser Publishers, 2011		
4. Vaezi, M., <i>3D Printing in Biomedical Engineering</i> , Springer, 2019		

2303BM034	FORENSIC SCIENCE IN HEALTHCARE					L	T	P	C			
	(B.E-BME)					3	0	0	3			
PREREQUISITE:												
	Human anatomy											
COURSE OBJECTIVES:												
	1.To introduce the fundamental principles and applications of forensic science in healthcare.											
	2.To understand medico-legal procedures, documentation, and forensic pathology											
	3.To analyze forensic techniques used in DNA profiling, toxicology, and digital evidence											
	4.To explore ethical, legal, and professional responsibilities in forensic investigations.											
COURSE OUTCOMES:												
On the successful completion of the course the students will be able to,												
CO1	Understand the role of forensic science in the medical and healthcare domain.											
CO2	Apply forensic methods for identification, documentation, and analysis of biological evidence.											
CO3	Analyze toxicological and pathological findings in medico-legal cases.											
CO4	Integrate DNA and digital forensic technologies in health-related investigations.											
CO5	Evaluate ethical, legal, and professional implications in forensic healthcare.											
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2									1	1
CO2	3	2	2				1				1	1
CO3	3	3	3	2							1	1
CO4	2	3	2	2	3	1					1	1
CO5											1	1
COs Vs PSOs MAPPING												
	COs	PSO1	PSO2	PSO3								
	CO1	2	1	2								
	CO2	2	2	2								
	CO3	2	2	2								
	CO4	2	2	3								
	CO5											
COURSE CONTENTS:												
MODULE I	INTRODUCTION TO FORENSIC SCIENCE AND MEDICO-LEGAL CONCEPTS								9 Hours			
History and scope of forensic science in healthcare. Structure and function of forensic labs. Medico-legal responsibilities of healthcare professionals. Cause, manner, and mechanism of death. Legal procedures: FIR, inquest, postmortem report, chain of custody.												
MODULE II	FORENSIC PATHOLOGY AND AUTOPSY TECHNIQUES								9 Hours			
Types of wounds and injuries – mechanical, thermal, electrical. Asphyxial deaths: Hanging, drowning, strangulation. Autopsy techniques: Clinical vs. forensic.Estimation of time since death. Forensic odontology and skeletal analysis.												

MODULE III	DNA PROFILING AND BIOMETRIC IDENTIFICATION	9 Hours
Collection and preservation of biological samples. DNA isolation, amplification (PCR), electrophoresis. Short Tandem Repeats (STRs), mtDNA, Y-STR analysis. Biometric tools: Fingerprint, iris, facial recognition in healthcare. Applications in paternity testing, disaster victim identification.		
MODULE IV	FORENSIC TOXICOLOGY AND CHEMICAL ANALYSIS	9 Hours
Introduction to poisons – classification, symptoms, and diagnosis. Analytical methods: Chromatography, spectrometry. Alcohol and drug analysis – in clinical and postmortem settings. Detection of toxins in biological samples. Role of toxicology in clinical trials and drug abuse cases.		
MODULE V	DIGITAL FORENSICS AND ETHICAL CONSIDERATIONS	9 Hours
Electronic health records (EHR) and digital evidence. Cybercrime in healthcare: Data breaches, ransomware, telemedicine risks. Digital autopsy and virtual postmortem. Ethical and legal issues in forensic practice. Patient rights, informed consent, and confidentiality in forensic settings.		
TOTAL:		45 Hours
FURTHER READING: Forensic Medicine and Toxicology		
Text books:		
1. B.S. Nabar, Forensic Science in Criminal Investigation and Trials, Asia Law House, 2015		
2. Gautam Biswas, Review of Forensic Medicine and Toxicology, Jaypee Brothers, 2018.		
References:		
3. Richard Saferstein, Criminalistics: An Introduction to Forensic Science, Pearson, 2015.		
4. William J. Tilstone, Kathleen M. Savage, Forensic Science: An Encyclopedia of History, Methods, and Techniques, ABC-CLIO, 2006.		
5. W. J. Chisum & B. E. Turvey, Crime Reconstruction, Academic Press, 2011.		

2303BM035		HAPTICS								L	T	P	C																																																																														
		(B.E-BME)								3	0	0	3																																																																														
PREREQUISITE:																																																																																											
		Anatomy and physiology, Bio sensor and transducer																																																																																									
COURSE OBJECTIVES:																																																																																											
		1.To introduce the fundamental concepts and human factors in haptic systems.																																																																																									
		2.To explore the design and working principles of haptic interfaces and devices.																																																																																									
		3.To analyze applications of haptics in biomedical simulation and rehabilitation.																																																																																									
		4.To understand control systems and feedback mechanisms in haptic interaction																																																																																									
COURSE OUTCOMES:																																																																																											
On the successful completion of the course the students will be able to,																																																																																											
CO1		Explain the principles of human touch perception and haptic feedback mechanisms.																																																																																									
CO2		Analyze the components and working of haptic devices and interfaces.																																																																																									
CO3		Design basic haptic interaction models with tactile and force feedback.																																																																																									
CO4		Apply haptic systems in biomedical simulations, robotic surgery, and rehabilitation.																																																																																									
CO5		Evaluate control strategies and stability issues in haptic rendering.																																																																																									
COs Vs POs MAPPING:																																																																																											
<table><tr><td>COs</td><td>PO1</td><td>PO2</td><td>PO3</td><td>PO4</td><td>PO5</td><td>PO6</td><td>PO7</td><td>PO8</td><td>PO9</td><td>PO10</td><td>PO11</td><td>PO12</td></tr><tr><td>CO1</td><td>3</td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>1</td><td></td></tr><tr><td>CO2</td><td>3</td><td>2</td><td>2</td><td></td><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>CO3</td><td>2</td><td>3</td><td>3</td><td>2</td><td>3</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>CO4</td><td>2</td><td>3</td><td>3</td><td>2</td><td>3</td><td></td><td>1</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>CO5</td><td>2</td><td>3</td><td>2</td><td>3</td><td>3</td><td>1</td><td>2</td><td>1</td><td></td><td></td><td>1</td><td>2</td></tr></table>														COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	CO1	3	2									1		CO2	3	2	2		2								CO3	2	3	3	2	3								CO4	2	3	3	2	3		1						CO5	2	3	2	3	3	1	2	1			1	2
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12																																																																															
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COs	PSO1	PSO2	PSO3																																																																																								
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COURSE CONTENTS:																																																																																											
MODULE I		INTRODUCTION TO HAPTICS										9 Hours																																																																															
Definition, scope, and significance of haptics in biomedical engineering. Human touch perception: Mechanoreceptors, tactile acuity, and proprioception. Types of haptic interfaces – tactile, kinesthetic, and cutaneous. Comparison with other sensory modalities. Basic block diagram of a haptic system.																																																																																											
MODULE II		HAPTIC INTERFACES AND DEVICES										9 Hours																																																																															
Components of a haptic system: actuators, sensors, controller, human operator.Tactile display technologies – vibration motors, piezoelectric actuators, electroactive polymers.Force-feedback devices – PHANToM, Novint Falcon, Omega, exoskeletons.Degrees of freedom (DoF), workspace, and resolution.Characteristics of haptic devices: stiffness, damping, transparency.																																																																																											
MODULE III		HAPTIC INTERACTION MODELS										9 Hours																																																																															

Modeling of virtual environments. Contact detection and force computation. Haptic rendering algorithms: point-based, surface-based models. Tactile rendering strategies – textures, temperature, compliance. Control loop for haptic interaction – update rates and latency.		
MODULE IV	BIOMEDICAL APPLICATIONS OF HAPTICS	9 Hours
Haptics in surgical simulation and training. Haptic-enabled robotic surgery systems. Rehabilitation and assistive technology – stroke rehab, prosthetics. Dental simulation, palpation training, and teleoperation. Case studies on haptics in physical therapy and diagnostics.		
MODULE V	CONTROL, STABILITY, AND ADVANCED TOPICS	9 Hours
Control strategies – impedance vs admittance control. Stability criteria and time delay issues. Bilateral control and tele-haptics. Integration with VR/AR and wearable haptics. Trends: Haptics in brain computer interfaces, haptic internet, 4D haptics		
TOTAL:		45 Hours
FURTHER READING: Haptics for Teleoperation		
Text books:		
1. Blake Hannaford & Allison M. Okamura, <i>Haptics for Virtual Reality and Teleoperation</i> , Springer Handbook, 2016.		
2. Vincent Hayward, <i>Haptics: Science, Technology, and Applications</i> , Springer, 2017.		
References:		
1.K. Salisbury & W. J. Barbagli, <i>Haptic Rendering: Foundations, Algorithms, and Applications</i> , A K Peters, 2008.		
2.M. Tavakoli et al., <i>Haptics for Teleoperated Surgical Robotic Systems</i> , World Scientific, 2021		
3.M. O'Malley et al., <i>Introduction to Haptics</i> , CRC Press (to be supplemented with online tutorials and device manuals)		

2303BM036	COMPUTER VISION					L	T	P	C			
	(B.E-BME)					3	0	0	3			
PREREQUISITE:												
	Medical Imaging											
COURSE OBJECTIVES:												
	1.To introduce the principles and techniques of computer vision.											
	2.To explore image formation, preprocessing, segmentation, and feature extraction.											
	3.To apply computer vision in biomedical image analysis and diagnostics.											
	4.To understand machine learning integration for intelligent visual interpretation											
COURSE OUTCOMES:												
On the successful completion of the course the students will be able to,												
CO1	Explain the fundamentals of image formation and perception in computer vision systems.											
CO2	Apply image preprocessing and enhancement techniques for biomedical images.											
CO3	Implement feature extraction and image segmentation algorithms.											
CO4	Analyze and interpret visual data using classification and object recognition techniques.											
CO5	Apply computer vision algorithms in biomedical engineering applications.											
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2									1	1
CO2	3	3	2	2	2						1	1
CO3	3	3	3	2	3						1	1
CO4	2	3	3	3	3	1	1		1		1	1
CO5	2	2	2	2	3	2	2	1	1		1	1
COs Vs PSOs MAPPING												
COs				PSO1	PSO2	PSO3						
CO1				2	2	2						
CO2				2	2	2						
CO3				2	2	3						
CO4				2	3	3						
CO5				2	3	3						
COURSE CONTENTS:												
MODULE I		INTRODUCTION TO COMPUTER VISION							9 Hours			
Overview of computer vision – human vs machine vision. Applications in biomedical engineering, robotics, surveillance, diagnostics. Image formation, light, color spaces, pinhole camera model. Geometric transformations – translation, scaling, rotation. Digital image representation – pixel, resolution, aspect ratio.												
MODULE II		IMAGE PROCESSING AND ENHANCEMENT							9 Hours			
Image acquisition – biomedical imaging modalities.Point operations – contrast adjustment, histogram equalization. Filtering – smoothing, sharpening, Gaussian, median filters.Noise models – salt & pepper, Gaussian, Poisson noise. Edge detection – Sobel, Prewitt, Laplacian, Canny.												
MODULE III		FEATURE EXTRACTION AND SEGMENTATION							9 Hours			

Image thresholding – global, local, and Otsu’s method. Region growing, splitting and merging. Morphological operations – dilation, erosion, opening, closing. Feature extraction – contours, corners, blobs (Harris, SIFT, SURF). Texture analysis – GLCM, LBP. Shape descriptors.		
MODULE IV	MACHINE VISION AND OBJECT RECOGNITION	9 Hours
Pattern recognition basics – supervised and unsupervised learning. Object classification using SVM, K-NN, decision trees. Template matching and Hough transform. Face detection, motion tracking, optical flow. Introduction to deep learning models (CNNs) for image classification.		
MODULE V	APPLICATIONS IN BIOMEDICAL ENGINEERING	9 Hours
Hours image segmentation – MRI, CT, ultrasound. Detection of tumors, lesions, lung nodules, diabetic retinopathy. Computer-aided diagnosis (CAD) systems. Vision-based prosthetics and assistive technologies Real-time processing in smart healthcare systems and wearable devices.		
TOTAL:		45 Hours
FURTHER READING: Image processing machine vision		
Text books:		
1. Richard Szeliski, <i>Computer Vision: Algorithms and Applications</i> , Springer, 2010.		
2. Rafael C. Gonzalez and Richard E. Woods, <i>Digital Image Processing</i> , Pearson, 2018.		
References:		
3. Sonka, Hlavac, and Boyle, <i>Image Processing, Analysis, and Machine Vision</i> , Cengage, 2015.		
4. Simon Prince, <i>Computer Vision: Models, Learning, and Inference</i> , Cambridge University Press, 2012.		
5. D. Forsyth and J. Ponce, <i>Computer Vision: A Modern Approach</i> , Pearson, 2015.		

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	(B.E-BME)						3	0	0	3		
PREREQUISITE:												
	Bio informatics											
COURSE OBJECTIVES:												
	1.To understand the evolving field of bioinformatics.											
	2.To analyse large biological data sets											
	3.To formulate the usage of biological tools effectively											
COURSE OUTCOMES:												
On the successful completion of the course the students will be able to,												
CO1	Implement Bioinformatics in data generation											
CO2	Identify appropriate biological data base for various analysis											
CO3	Apply suitable analytical techniques to Biological data for research.											
CO4	Implement the concepts of genomic technology in bio informatics											
CO5	Analyse genetic variability of clinical data											
COs Vs POs MAPPING:												
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		2	2	3								
CO2		2	3	2								
CO3		2	3	2								
CO4		2	3	2								
CO5		2	1	3				1				
COs Vs PSOs MAPPING												
	COs	PSO1	PSO2	PSO3								
	CO1	1	2									
	CO2	1	2									
	CO3	1	2									
	CO4	1	2									
	CO5	1	2									
COURSE CONTENTS:												
MODULE I	INTRODUCTION TO BIOINFORMATICS AND DATA GENERATION										9 Hours	
Bioinformatics and its relation with molecular biology. Examples of tools (FASTA, BLAST, BLAT RASMOL), databases (GENBANK, PubMed, PDB), Data generation; Generation of large scale molecular biology data , Genome sequencing, Protein sequencing, Gel electrophoresis, NMR Spectroscopy, X, Ray Diffraction, and microarray, Applications of Bioinformatics												
MODULE II	BIOLOGICAL DATA BASE AND ITS TYPES										9 Hours	
General Introduction of Biological Databases; Nucleic acid databases (NCBI, DDBJ, and EMBL).Protein data bases (Primary, Composite and Secondary). Specialized Genome data												

bases: (SGD,TIGR, and ACeDB). Structure databases (CATH, SCOP, and PDB sum).		
MODULE III	STRUCTURAL BIOINFORMATICS	9 Hours
Fundamentals of X-ray diffraction,NMR spectroscopy of macromolecules, Protein Structure: Primary, Secondary, Super Secondary, Domains, Tertiary, Quaternary, Structural features of RNA: Primary, Secondary, Tertiary, Motif and Domain: Motif databases and analysis tools. Domain databases(CDD, SMART, ProDom) and Analysis tools.		
MODULE IV	DIFFERENT TYPES AND METABOLIC PATHWAYS	9 Hours
Genomics:GenomeAnnotation,GenomeAssembly,StructuralandFunctionalGenomics.Comparative Genomics, Metagenomics: Introduction, metagenome, shotgun metagenomics (pyrosequencing) Metabolic pathway database (KEGG pathway database), Concept of metabolome and metabolomics		
MODULE V	GENE EXPRESSION AND REPRESENTATION OF PATTERNS AND RELATIONSHIP	9 Hours
General introduction to Gene expression in prokaryotes and eukaryotes, transcription factors binding sites. SNP, EST, STS. Introduction to Regular Expression, Hierarchies, and Graphical models (including Markov chain and Bayes notes). Genetic variability and connections to clinical data.		
		TOTAL: 45 Hours
FURTHER READING: Application of medical informatics		
Text books:		
1.Introduction to Bioinformatics Algorithms by Neil Jones and Pavel Pevzner.2015.		
2.Bio informatics by David Mount,2016.		
References:		
1.Principles and Applications by Zhumur Ghosh and Bibek anand Mallick,2010.		
2.Bioinformatics:Sequence and Genome Analysis by Mount and David W,2005		